



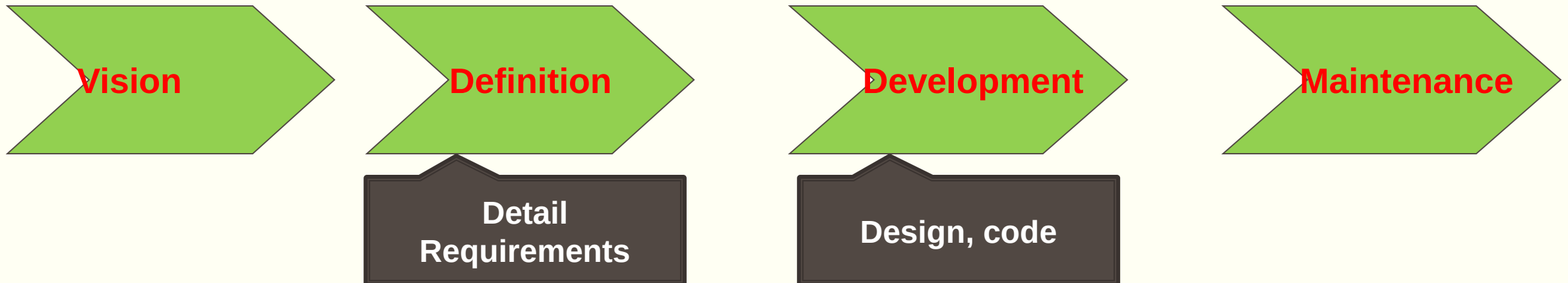
Software Engineering

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Software Engineering Phases

1. Vision – focus on *why*
2. Definition – focus on *what*
3. Development – focus on *how* (*how we can achieve this technically, design, code*)
4. Maintenance – focus on *change*



Generic Process Framework

- Communication
 - Involves communication among the customer and other stake holders; encompasses requirements gathering
- Planning
 - Establishes a plan for software engineering work; addresses technical tasks, resources, work products, and work schedule
- Modelling (Analyse, Design)
 - Encompasses the creation of models to better understand the requirements and the design

- Construction (Code, Test)

- Combines code generation and testing to uncover errors

- Deployment

- Involves delivery of software to the customer for evaluation and feedback

Modelling: Software Requirements Analysis

- Helps software engineers to better understand the problem they will work to solve
- Encompasses the set of tasks that lead to an understanding of what the business impact of the software will be, what the customer wants, and how end-users will interact with the software
- Uses a combination of text and diagrams to depict requirements for data, function, and behaviour
 - Provides a relatively easy way to understand and review requirements for correctness, completeness and consistency

Modelling: Software Design

- Brings together customer requirements, business needs, and technical considerations to form the “blueprint” for a product
- Creates a model that provides detail about software data structures, software architecture, interfaces, and components that are necessary to implement the system
- Architectural design
 - Represents the structure of data and program components that are required to build the software
 - Considers the architectural style, the structure and properties of components that constitute the system, and interrelationships that occur among all architectural components

- User Interface Design

- Creates an effective communication medium between a human and a computer
- Identifies interface objects and actions and then creates a screen layout that forms the basis for a user interface prototype

- Component-level Design

- Defines the data structures, algorithms, interface characteristics, and communication mechanisms allocated to each software component

Embracing Change

- In software projects one thing that is constant:



What is Software Process?

- *“A software process is a set of related activities that leads to the production of a software product ”*



Software Processes Fundamentals Activities

Software Specification:

The functionality of the software and constraints on its operation must be defined

Software Design and Implementation:

The software to meet the specification must be produced

Software Processes Fundamentals Activities

Software Validation:

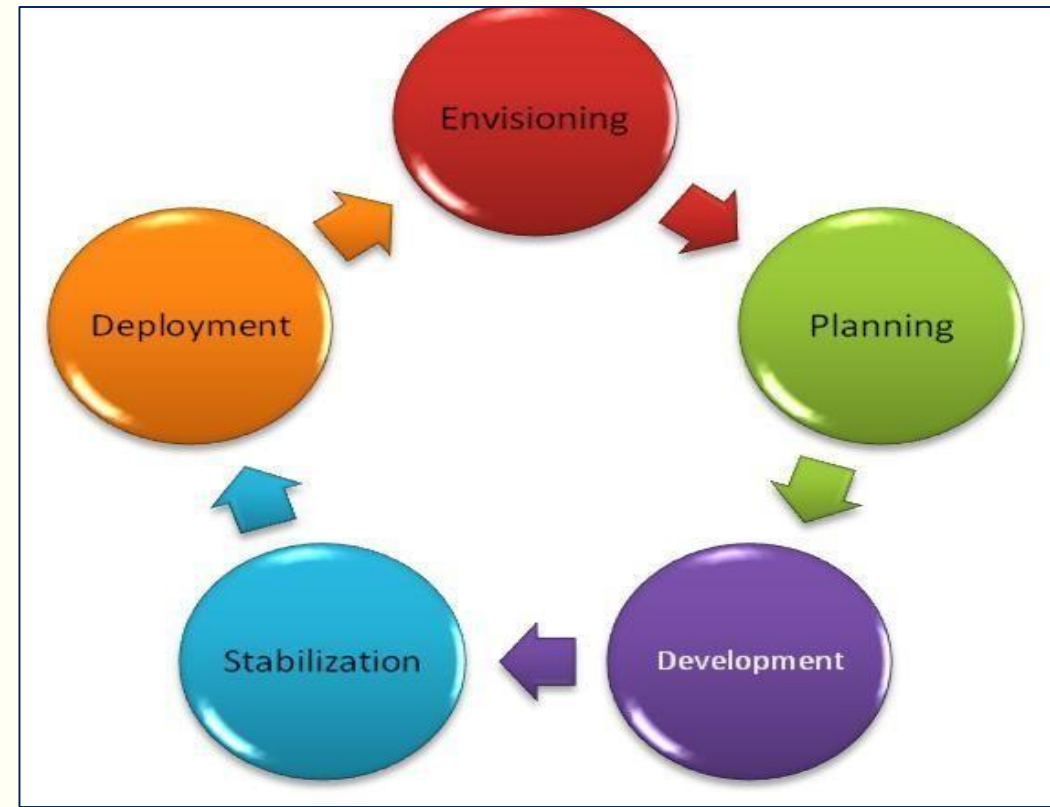
The software must be validated to ensure that **it does** what the **customer wants**

Software Evolution:

The software must evolve to meet changing customer needs

Software Process Models

- A software process model is a simplified representation of a software process
- Each process model represents a process from a particular perspective
- Provides only partial information about that process



Process Model

- Defines a distinct **set of activities, actions, tasks, milestones, and work products** that are required to engineer high-quality software
- The activities may be linear, incremental, or evolutionary

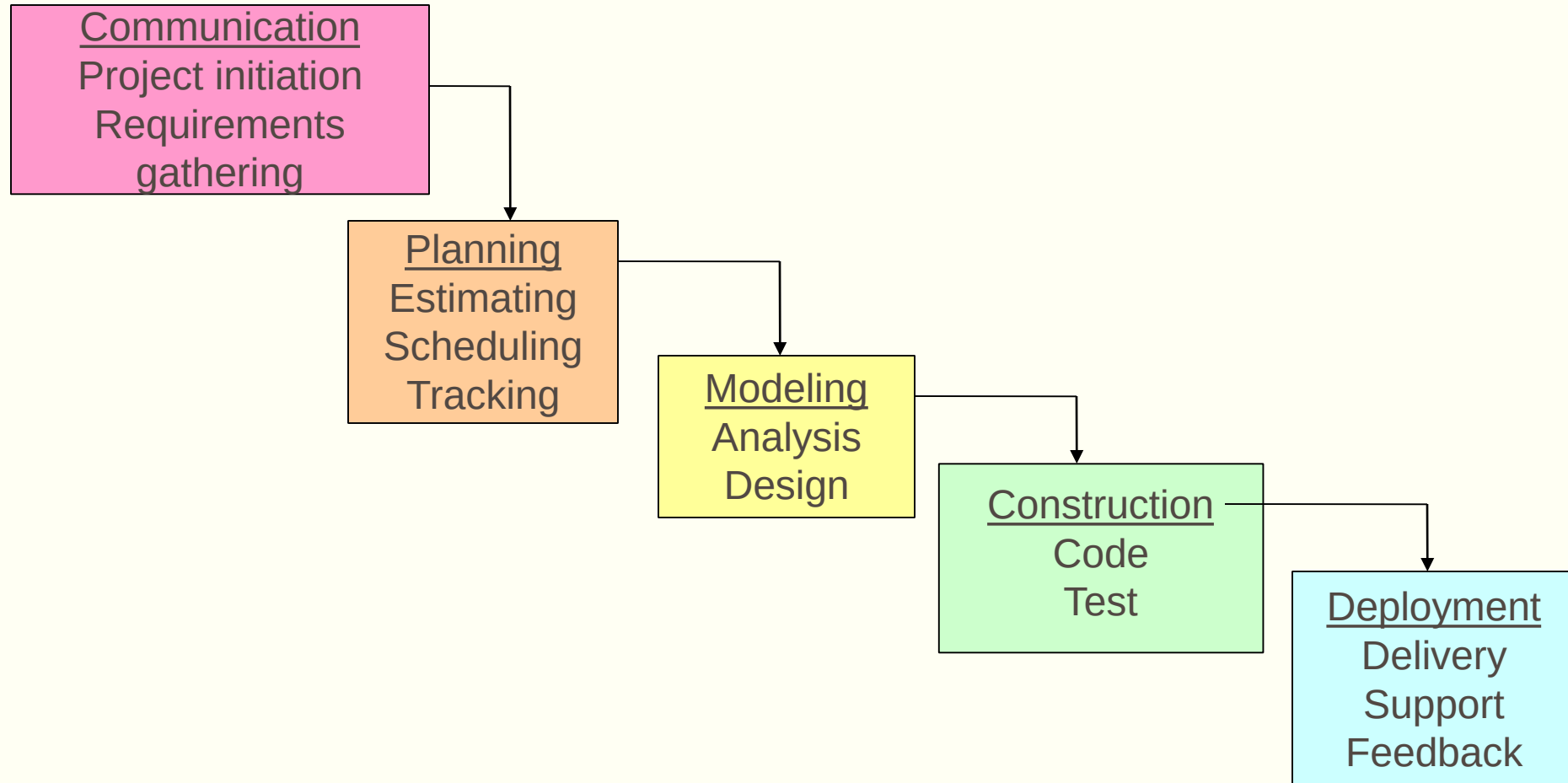


The Waterfall Model

- The first published model of the software development process (Royce, 1970)
- Because of the cascade from one phase to another, this model is known as the 'Waterfall Model'
- **Plan and schedule all of the process activities before starting work on them**



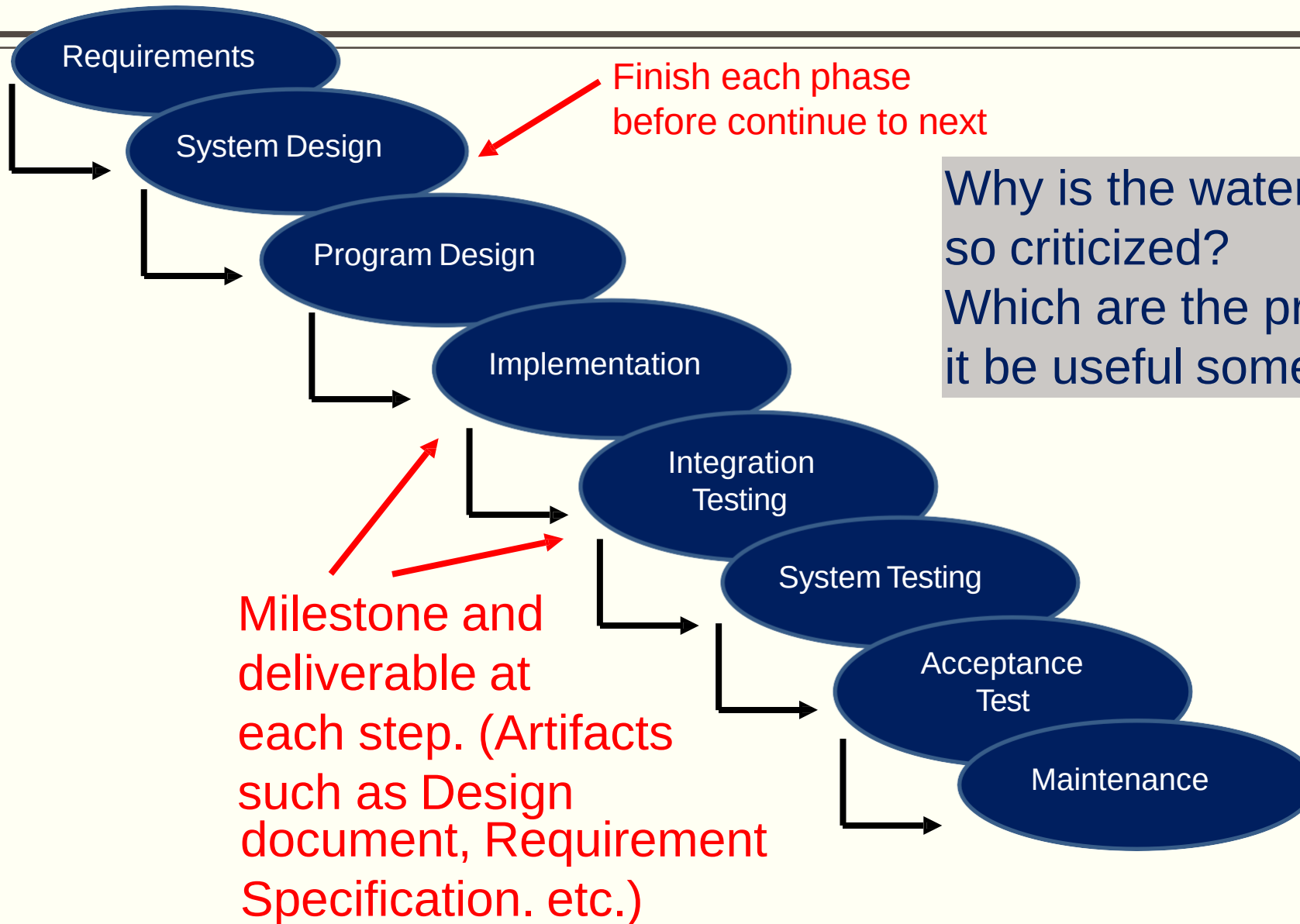
Waterfall Model (Diagram)



Waterfall model problems

- ✧ Inflexible partitioning of the project into distinct stages makes it difficult to respond to changing customer requirements.
 - Therefore, this model is only appropriate when the requirements are well-understood and changes will be fairly limited during the design process.
 - Few business systems have stable requirements.
- ✧ The waterfall model is mostly used for large system engineering projects where a system is developed at several sites.
 - In those circumstances, the plan-driven nature of the waterfall model helps coordinate the work.

The Waterfall Model



Why is the waterfall model so criticized?
Which are the problems? Can it be useful sometimes?